Working methodology (token economy)

The whole conversation is re-sent on every turn, so the method is: never load geometry into chat, verify by numbers, keep turns few, and let Jordane's own FreeCAD do the looking.

Rules

1. Never read a CAD file into the conversation. Query it with a short script that prints a one-line summary. (60 MB STEP ≈ 20M tokens; `Document.xml` ≈ 3M; the `.kicad_pcb` ≈ 30k.)
2. Upload big files as attachments, not `@`-mentions; `@` reads the file into context and it rides along every turn after.
3. Verify numerically (bounding boxes, distances, volumes). A screenshot costs 1,300–2,700 tokens (⌈w/28⌉ × ⌈h/28⌉); Jordane looking in FreeCAD costs nothing.
4. Fewer, bigger turns. One script that does the whole step beats ten incremental runs.
5. Separate conversations per project; this one carries a large context.
6. Max effort for design decisions; lower effort for "run it again" turns.
7. Scripts are minimal and hardcoded to this project; no generalizing.
8. Code is written in reviewed pieces so a person can read every line.

Tooling in the sandbox

- **All generated code runs on Jordane's Mac, in his FreeCAD (macro editor).** Claude writes the piece; Jordane runs it and pastes back the printed report or the error. This keeps colors, views and file paths in the real environment and costs ~1–2k tokens per round.
- The sandbox has FreeCAD 1.1.0 headless (`freecadcmd`) for syntax/logic pre-checks only; it has no GUI, so it cannot set colors, and a hand-written GuiDocument.xml crashes FreeCAD on open (learned 2026-10-09). Never ship an FCStd from the sandbox.
- Board file parsed with plain Python. KiCad itself is not installable in the sandbox (only v7 reachable; file is v10.99).
- Available via pip if phase 2 needs them: `trimesh`, `manifold3d` (mesh edits), `sexpdata` / `kiutils` (KiCad parsing).

Phase 2 paths, ranked by cost (for later)

Path	Cost	When
Hybrid: vendor solid kept; base = solid ∩ box below `z_split`, decoration above; parametric cuts on the base	~20–40k tokens	default
Mesh: STL once, `trimesh` + `manifold3d` edits	~10–20k	quick print tests
Full parametric remake	~80–150k, uncertain	only if the top surface itself must change, and only after checking for an existing parametric source

Sources

- Context windows · Vision token cost
- FreeCAD 1.1.0 release · Headless FreeCAD · Std Part · Expressions · Std LinkMake · Part scripting · Topological data scripting · Part Common · Part Cut
- KiCad PCB file format · kicad-cli
- trimesh · manifold
- Sprintek DS0048 SK8707-01 datasheet ver. 1.04 (attached to the project chat) · DS0046 SK8707 datasheet